

Hugging Face Hub and Open-Source Models

The Hugging Face Ecosystem Hugging Face is the central platform for open-source AI. It hosts models, datasets, and interactive spaces. The `transformers` library provides APIs to download, load, and run state-of-the-art open models locally.

Sentence Transformers and Embeddings Sentence Transformers is a framework to generate high-quality vector representations (embeddings) of sentences, paragraphs, and images. The model `all-MiniLM-L6-v2` is a lightweight, widely-used transformer model mapping text to 384-dimensional vectors, ideal for local semantic search.

Tokenization Tokenizers convert raw text strings into numerical tokens. Byte-Pair Encoding (BPE) and WordPiece are common tokenization algorithms that segment words to handle out-of-vocabulary terms.